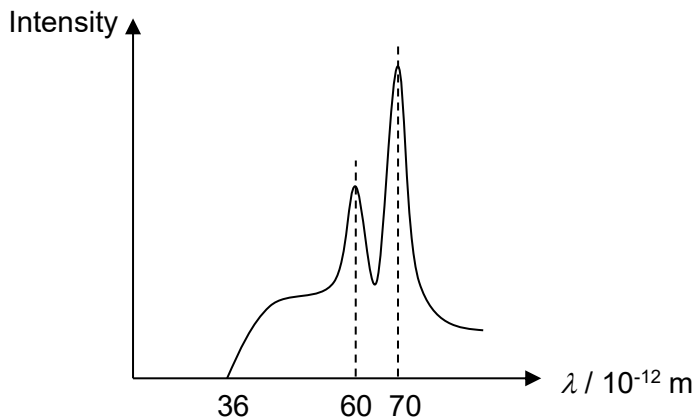


28 The x-ray spectrum produced by bombarding a metal target with electrons is shown below.



Which of the following statements is correct?

- A The graph shows that electrons with a range of kinetic energies are used to bombard the target.
- B The wavelength corresponding to the peaks allow the energy of the electrons used to bombard the target to be calculated.
- C The smallest wavelength detected, $36 \times 10^{-12} \text{ m}$, is dependent on the target material.
- D The wavelength corresponding to the peaks can be used to identify the element of target.

Ans: D

Option A is not correct because since the X-ray spectra will always contain bremsstrahlung radiation whether the electrons bombarding the metal target has a single energy or have a range of energies. Option B is not correct because the the position (wavelength) corresponding to the characteristic peaks is dependent only on the kind of target and not on the energy of the incoming electrons. Option C is not correct because minimum wavelength (cutoff wavelength) depends on the maximum energy of the incident electrons.

29 The fusion of two deuterium nuclei produces a nuclide of helium and a neutron. The reaction